JIEMING ZHANG

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EDUCATION

University of Virginia **Charlottesville, VA**

B.S. Computer Engineering & B.S. Electrical Engineering, ***GPA : 3.9*** *Expected Graduation: May 2029*

Relevant Coursework: Computer Systems Organization, Electric Circuits, Digital Logic Design, CPU Architecture, Ordinary Differential Equations, Multivariable Calculus, Data Structures & C++

TECHNICAL SKILLS

Languages: C++, Python, VHDL, Java

Hardware & EDA: Custom PCB Design, Power Management (3.7V LiPo to 3.3V LDO), KiCad, Altium, Quartus, ModelSim, ESP32, BNO055 IMU, MicroSD SPI, Multimeters

Protocols & Tools: I2C, SPI, UART, CAN Bus, Fusion 360, Git, Matplotlib, SciPy

Concepts: Low-Level Firmware, Digital Logic, CPU Architecture, Signal Processing, Sensor Fusion

ENGINEERING EXPERIENCE

Virginia Motorsports (Formula SAE) **Fall 2025 - Present**

Data Acquisition (DAQ) Subteam Member | University of Virginia

- Contributing to a redesign consolidating 5 discrete PCBs into a single centralized sensorboard handling 17 analog sensor inputs across 4 sensor types via Teensy, cutting wiring complexity and points of failure.
- Design schematics and multi-layer PCBs in Altium and KiCad for sensor integration and data logging on an all-electric FSAE racecar.
- Implementing SD card logging by accessing sensor readings directly from the shared DAQ struct, writing synchronized data across all 17 sensor channels to persistent storage.

TECHNICAL PROJECTS

IoT Golf Swing Analyzer & Visualizer Prototype **Summer 2026**

Embedded Hardware, Firmware & Python Analytics · Personal Project

- Designed and routed a custom ESP32 PCB (~$20 in components) with 3.7V LiPo-to-3.3V power management and interfaces for the IMU and MicroSD logger.
- Built non-blocking C++ firmware with button-triggered record/idle state control to log 100 Hz BNO055 IMU quaternion and accelerometer data over I2C, writing CSV output to SPI MicroSD storage.
- Created a Python telemetry engine using Matplotlib and SciPy to process raw quaternions, correct gravitational drift, estimate clubhead velocity, and render 3D swing trajectories.

CPU Datapath & Controller Design **Spring 2026**

UVA ECE Coursework

- Architected a structural VHDL CPU datapath with ALU functions, register banks, and control sequencers.
- Debugged timing and control-state faults in ModelSim using testbench reset pulses and waveform traces; synthesized the design in Intel Quartus.

Automated Physical Sink Shutoff System **Fall 2025**

Introductory Engineering Design Project · UVA

- Engineered a retrofit automated shutoff mechanism for standard sink handles without internal plumbing access.
- Integrated an ultrasonic distance sensor to detect user presence and trigger automated shutoff sequences.
- Designed and calibrated a servo-driven mechanical arm, triggered by ultrasonic distance readings detecting user movement, to actuate the external water valve handle.

HACKATHON EXPERIENCE

VTHacks 14 **Sep 2026**

2-person team · Embedded Systems & Wireless Sensing

- Built an ESP32 occupancy-sensing prototype that scanned 2.4 GHz Wi-Fi and Bluetooth traffic for packet counts and RSSI.
- Developed C++ firmware to compute a 0-100 density score from signal activity and fluctuation, calibrated against an empty-room baseline stored in nonvolatile memory.
- Produced device measurements for a partner-developed Supabase and React/Node.js dashboard that visualized occupancy trends.